

Monoidal coherence of the sequential operator and the Dirichlet tensor, in explicit container language

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Abstract

We supply self-contained, container-level verifications of the monoidal coherence that the “four monoidal structures” chapter of the book currently cites from Spivak. Part (A) treats the *sequential operator* $(\mathbf{Cont}, \triangleleft, y)$: we write the associator α and the unitors λ, ρ as explicit shape/position bijections and verify the *pentagon* and *triangle* identities as exact equalities of those bijections. The associator’s shape component is *currying* and its position component is Σ -*reassociation*; both pentagon legs are shown to equal a single *flattening* of a four-fold composite to its parenthesization-free normal form, and on positions every map in sight is the identity on the underlying flat tuple, so coherence is forced. Part (B) treats the *Dirichlet tensor* $(\mathbf{Cont}, \otimes, y)$: we give the comparison $\varphi_{C_1, C_2} : \llbracket C_1 \otimes C_2 \rrbracket \xrightarrow{\sim} \llbracket C_1 \rrbracket \otimes_{\text{Dir}} \llbracket C_2 \rrbracket$, show it is the identity once $\otimes_{\text{Dir}}$ is taken in its native polynomial presentation (so $\llbracket - \rrbracket$ is in fact *strict* monoidal here, with $\llbracket y \rrbracket = \text{Id}$), and reduce its associativity coherence to Cartesian reassociation of shape and position sets. [REFUTED 2026-07-14 — see Erratum: $\llbracket - \rrbracket$ is *strong*, not strict.] We isolate the genuine subtlety of $\otimes$: unlike $+$, $\times$, $\triangleleft$ its target is *not* a pointwise/compositional operation on the underlying functors but depends on the polynomial presentation. [The diagnosis is REFUTED 2026-07-14 — see Erratum: $\times$ is a Day convolution too, and is pointwise.] Every identity below was also confirmed by exhaustive machine computation on small containers (120 randomised four-fold instances plus a branching stress case).

Erratum (2026-07-14). Two claims of Part (B) are refuted. I state them plainly.

(1) φ is *not* the identity and $\llbracket - \rrbracket$ is *not* strict monoidal for $\otimes$ (Theorem 3 is false as stated). The argument is a category error: I *defined* $\otimes_{\text{Dir}}$ by reading off the container presentation (S, P) — I transported the structure along $\llbracket - \rrbracket$ and then observed that $\llbracket - \rrbracket$ preserves it. That is circular. Intrinsically, the Dirichlet tensor is the *Day convolution* on $[\mathbf{Set}, \mathbf{Set}]$ with respect to $(\mathbf{Set}, \times, 1)$, and the comparison map is the co-Yoneda/density *isomorphism* out of a coend, not an identity. The correct statement: $\llbracket - \rrbracket$ is **strong** monoidal for $\otimes$, not strict.

(2) “ $\otimes$ is not pointwise *because it is a Day convolution*” is false (Remark 2). The product $\times$ on $\mathbf{Cont}$ is *also* a Day convolution — of $(\mathbf{Set}, +, 0)$ — and it *is* pointwise. The correct criterion is whether the corepresentable splits the domain tensor: $y^{A+B} \cong y^A \times y^B$ (the exponential law), so Day-of- $+$ is pointwise; $y^{A \times B}$ admits no such splitting, so Day-of- $\times$ is not. Witness: $y^3 \otimes y^3 = y^9$, against the pointwise y^6 .

Attribution. The identification “Dirichlet tensor = Day convolution” is *not mine*. It is Niu–Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, arXiv:2312.00990, Proposition 3.79 (pp. 69–71), with essentially the same proof: “For any monoidal structure $(I, \star)$ on $\mathbf{Set}$, there is a corresponding monoidal structure $(y^I, \odot)$ on $\mathbf{Poly}$, where $\odot$ is the Day convolution. ... In the case of $(0, +)$ and $(1, \times)$, this procedure returns the $(1, \times)$ and

$(y, \otimes)$ monoidal structures respectively.” The species analogue is the Maia–Méndez arithmetic product (arXiv:math/0503436).

What survives. Part (A) is untouched: the explicit associator and unitors for $\triangleleft$, the pentagon (Theorem 1) and triangle (Theorem 2), and the Lean instance `ContMonoidal : MonoidalCategory Container` that certifies them. The refutation is confined to the Dirichlet strictness sub-claim and the reasoning about pointwiseness. The erroneous text below is retained and marked, not deleted: the record of the error has value.

1 Conventions

A *container* is a pair $C = (S, P)$ with $S : \mathbf{Set}$ a set of *shapes* and $P : S \rightarrow \mathbf{Set}$ a family of *position* sets; we write $C = S \triangleleft P$. Its extension is $\llbracket S \triangleleft P \rrbracket X = \sum_{s:S} (P s \rightarrow X)$.

A morphism of containers $\mu : A \rightarrow B$ is a pair (μ_s, μ_p) where $\mu_s : \text{Sh } A \rightarrow \text{Sh } B$ on shapes and, *contravariantly on positions*, a family $\mu_{p,a} : \text{Pos}_B(\mu_s a) \rightarrow \text{Pos}_A(a)$. It is an isomorphism iff μ_s and every $\mu_{p,a}$ are bijections. This is the standard convention under which $\llbracket - \rrbracket : \mathbf{Cont} \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is fully faithful (Abbott–Altenkirch–Ghani).

Throughout, $C_i = (S_i, P_i)$, and $y = 1 \triangleleft 1$ is the container with one shape $*$ and one position $\bullet$, so $\llbracket y \rrbracket = \text{Id}$.

2 Part (A): the sequential operator $(\mathbf{Cont}, \triangleleft, y)$

2.1 The operation and its unit

Definition 1 (Sequential operator).

$$C_1 \triangleleft C_2 = \left(\sum_{s_1:S_1} (P_1 s_1 \rightarrow S_2) \right) \triangleleft ((s_1, f) \mapsto \sum_{p_1:P_1 s_1} P_2(f p_1)).$$

A shape is a pair (s_1, f) with $f : P_1 s_1 \rightarrow S_2$; a position over (s_1, f) is a pair (p_1, p_2) with $p_1 : P_1 s_1$ and $p_2 : P_2(f p_1)$.

This realises composition of extensions: $\llbracket C_1 \triangleleft C_2 \rrbracket = \llbracket C_1 \rrbracket \circ \llbracket C_2 \rrbracket$ (verified in `Composition.lean`). The unit is y , since $y \triangleleft C \cong C \cong C \triangleleft y$ (Section 2.3).

2.2 The associator

Proposition 1 (Explicit associator). *There is an isomorphism $\alpha = \alpha_{C_1, C_2, C_3} : (C_1 \triangleleft C_2) \triangleleft C_3 \xrightarrow{\sim} C_1 \triangleleft (C_2 \triangleleft C_3)$ given as follows.*

A shape of $(C_1 \triangleleft C_2) \triangleleft C_3$ is $((s_1, f), g)$ with $f : P_1 s_1 \rightarrow S_2$ and $g : (\sum_{p_1} P_2(f p_1)) \rightarrow S_3$. A shape of $C_1 \triangleleft (C_2 \triangleleft C_3)$ is (s_1, k) with $k : P_1 s_1 \rightarrow \text{Sh}(C_2 \triangleleft C_3)$. On shapes,

$$\alpha_s((s_1, f), g) = (s_1, k), \quad k p_1 = (f p_1, \lambda p_2. g(p_1, p_2)).$$

That is, α_s is the currying bijection $\sum_{f:P_1 s_1 \rightarrow S_2} (\sum_{p_1} P_2(f p_1) \rightarrow S_3) \cong (P_1 s_1 \rightarrow \sum_{s_2} (P_2 s_2 \rightarrow S_3))$.

On positions, both sides have the set $\sum_{p_1} \sum_{p_2:P_2(f p_1)} P_3(g(p_1, p_2))$, written $\{(p_1, p_2), p_3\}$ on the left and $\{(p_1, (p_2, p_3))\}$ on the right, and

$$\alpha_p : (p_1, (p_2, p_3)) \mapsto ((p_1, p_2), p_3)$$

is the Σ -reassociation bijection.

Proof. α_s is the composite of the type-theoretic distributivity equivalence $\prod_{p_1} \sum_{s_2} (P_2 s_2 \rightarrow S_3) \cong \sum_f \prod_{p_1} (P_2(f p_1) \rightarrow S_3)$ and the currying equivalence $\prod_{p_1} (P_2(f p_1) \rightarrow S_3) \cong (\sum_{p_1} P_2(f p_1) \rightarrow S_3)$, both bijections; its inverse sends (s_1, k) to $((s_1, f), g)$ with $f p_1 = \pi_1(k p_1)$ and $g(p_1, p_2) = \pi_2(k p_1) p_2$. After substituting $k p_1 = (f p_1, \lambda_{p_2}.g(p_1, p_2))$ the two position sets are literally the same iterated sum, reassociated; α_p is the canonical bijection $\sum_{(p_1, p_2): \sum_{p_1} P_2(f p_1)} P_3(g(p_1, p_2)) \cong \sum_{p_1} \sum_{p_2} P_3(g(p_1, p_2))$. Both components are bijections, so α is an isomorphism. $\square$

Remark 1. Both components are “transport-free”: the shape map is currying and the position map is Σ -reassociation, matching the transport-free associator in `Composition.lean` (PR #13).

2.3 The unitors

Proposition 2. $\lambda_C: y \triangleleft C \xrightarrow{\sim} C$ and $\rho_C: C \triangleleft y \xrightarrow{\sim} C$ are:

$$\lambda_s(*, g) = g \bullet, \quad \lambda_p: p \mapsto (\bullet, p); \quad \rho_s(s, !) = s, \quad \rho_p: p \mapsto (p, \bullet).$$

Proof. A shape of $y \triangleleft C$ is $(*, g)$ with $g: 1 \rightarrow S$, i.e. an element $g \bullet \in S$; its positions are $\sum_{p:1} P(g \bullet) \cong P(g \bullet)$. A shape of $C \triangleleft y$ is $(s, !)$ with $!: Ps \rightarrow 1$ unique, i.e. an element $s \in S$; its positions are $\sum_{p:Ps} 1 \cong Ps$. Both maps are bijections on shapes with the stated position bijections. $\square$

2.4 Whiskering

We record the action of $\triangleleft$ on morphisms (its functoriality), needed for the pentagon’s whiskered edges. For $\mu: A \rightarrow A'$,

$$(\mu \triangleleft C)_s(a, h) = (\mu_s a, h \circ \mu_{p,a}), \quad (\mu \triangleleft C)_p(a, h): (y, p) \mapsto (\mu_{p,a} y, p);$$

and for $\mu: B \rightarrow B'$,

$$(C \triangleleft \mu)_s(s_1, k) = (s_1, \mu_s \circ k), \quad (C \triangleleft \mu)_p(s_1, k): (p_1, y) \mapsto (p_1, \mu_p y).$$

2.5 The pentagon

Theorem 1 (Pentagon). *For all C_1, C_2, C_3, C_4 the diagram*

$$\begin{array}{ccc} ((C_1 \triangleleft C_2) \triangleleft C_3) \triangleleft C_4 & \xrightarrow{\alpha_{123 \triangleleft C_4}} & (C_1 \triangleleft (C_2 \triangleleft C_3)) \triangleleft C_4 \\ \downarrow \alpha_{12,3,4} & & \downarrow \alpha_{1,23,4} \\ (C_1 \triangleleft C_2) \triangleleft (C_3 \triangleleft C_4) & & C_1 \triangleleft ((C_2 \triangleleft C_3) \triangleleft C_4) \\ \downarrow \alpha_{1,2,34} & & \downarrow C_1 \triangleleft \alpha_{234} \\ C_1 \triangleleft (C_2 \triangleleft (C_3 \triangleleft C_4)) & = & C_1 \triangleleft (C_2 \triangleleft (C_3 \triangleleft C_4)) \end{array}$$

commutes: writing $a = \alpha_{123} \triangleleft C_4$, $b = \alpha_{1,23,4}$, $c = C_1 \triangleleft \alpha_{234}$, $d = \alpha_{12,3,4}$, $e = \alpha_{1,2,34}$, we have $c \circ b \circ a = e \circ d$.

Proof. Shapes. A shape of the source $((C_1 \triangleleft C_2) \triangleleft C_3) \triangleleft C_4$ is, by Definition 1 applied thrice, a tuple

$$W = (((s_1, f), g), j), \quad f: P_1 s_1 \rightarrow S_2, \quad g: \{(p_1, p_2)\} \rightarrow S_3, \quad j: \{((p_1, p_2), p_3)\} \rightarrow S_4,$$

where $\{(p_1, p_2)\} = \sum_{p_1} P_2(f p_1)$ and $\{((p_1, p_2), p_3)\} = \sum_{(p_1, p_2)} P_3(g(p_1, p_2))$. A shape of the target $C_1 \triangleleft (C_2 \triangleleft (C_3 \triangleleft C_4))$ is (s_1, K) with $K: P_1 s_1 \rightarrow \text{Sh}(C_2 \triangleleft (C_3 \triangleleft C_4))$. Define the *flattening*

$$\Phi(W) = (s_1, K), \quad K p_1 = \left(f p_1, \lambda_{p_2}.(g(p_1, p_2), \lambda_{p_3}.j((p_1, p_2), p_3)) \right). \quad (1)$$

We compute both legs and show each equals Φ .

Leg $c \circ b \circ a$. Using Section 2.4 for a and Proposition 1 for b, c :

$$\begin{aligned} a(W) &= ((s_1, k), j \circ \mu_p), \quad k p_1 = (f p_1, \lambda p_2 \cdot g(p_1, p_2)), \quad (j \circ \mu_p)(p_1, (p_2, p_3)) = j((p_1, p_2), p_3), \\ b(a(W)) &= (s_1, k'), \quad k' p_1 = (k p_1, \lambda(p_2, p_3) \cdot j((p_1, p_2), p_3)), \\ c(b(a(W))) &= (s_1, \lambda p_1 \cdot \alpha_{234}(k' p_1)), \quad \alpha_{234}(k' p_1) = (f p_1, \lambda p_2 \cdot (g(p_1, p_2), \lambda p_3 \cdot j((p_1, p_2), p_3))), \end{aligned}$$

where the last line applies α_{234} to the shape $((f p_1, \lambda p_2 \cdot g(p_1, p_2)), \lambda(p_2, p_3) \cdot j((p_1, p_2), p_3))$ of $(C_2 \triangleleft C_3) \triangleleft C_4$. Hence $c \circ b \circ a(W) = (s_1, K) = \Phi(W)$.

Leg $e \circ d$. Using Proposition 1 for both:

$$\begin{aligned} d(W) &= ((s_1, f), \kappa), \quad \kappa(p_1, p_2) = (g(p_1, p_2), \lambda p_3 \cdot j((p_1, p_2), p_3)), \\ e(d(W)) &= (s_1, \lambda p_1 \cdot (f p_1, \lambda p_2 \cdot \kappa(p_1, p_2))) = (s_1, K) = \Phi(W). \end{aligned}$$

So both legs agree on shapes.

Positions. Every position component above is, by Proposition 1 and Section 2.4, a bijection that is the *identity on the underlying flat tuple* (p_1, p_2, p_3, p_4) and only changes its bracketing: α_p rebrackets one adjacent pair, and a whisker leaves the whiskered coordinate fixed while transporting the rest. The source position set is bracketed $\{(((p_1, p_2), p_3), p_4)\}$ and the target $\{(p_1, (p_2, (p_3, p_4)))\}$; both are in canonical bijection with the flat set $\{(p_1, p_2, p_3, p_4)\}$ via the evident “erase brackets” map, call them $\text{fl}_{\text{src}}, \text{fl}_{\text{tgt}}$. Each elementary map m satisfies $\text{fl} \circ m_p = \text{fl}$, hence so does any composite; therefore both legs equal $\text{fl}_{\text{tgt}}^{-1} \circ \text{fl}_{\text{src}}$, the unique order-preserving rebracketing. They coincide. $\square$

2.6 The triangle

Theorem 2 (Triangle). *For all C_1, C_2 ,*

$$(C_1 \triangleleft \lambda_{C_2}) \circ \alpha_{C_1, y, C_2} = \rho_{C_1} \triangleleft C_2 : (C_1 \triangleleft y) \triangleleft C_2 \longrightarrow C_1 \triangleleft C_2.$$

Proof. A shape of $(C_1 \triangleleft y) \triangleleft C_2$ is $((s_1, f), g)$ with $f: P_1 s_1 \rightarrow 1$ (unique) and $g: \{(p_1, \bullet)\} \rightarrow S_2$, where $\{(p_1, \bullet)\} = \sum_{p_1} 1 \cong P_1 s_1$.

Left side. α_{C_1, y, C_2} gives $(s_1, k), k p_1 = (*, \lambda \bullet \cdot g(p_1, \bullet))$, a shape of $C_1 \triangleleft (y \triangleleft C_2)$. Whiskering by λ_{C_2} applies λ_{C_2} to each $k p_1$: $\lambda_{C_2}(*, \lambda \bullet \cdot g(p_1, \bullet)) = g(p_1, \bullet)$. The result is $(s_1, \lambda p_1 \cdot g(p_1, \bullet))$.

Right side. ρ_{C_1} sends $(s_1, f) \mapsto s_1$ with $\rho_p: p_1 \mapsto (p_1, \bullet)$; whiskering by C_2 (Section 2.4) gives $(s_1, g \circ \rho_p)$ with $(g \circ \rho_p)(p_1) = g(p_1, \bullet)$, i.e. $(s_1, \lambda p_1 \cdot g(p_1, \bullet))$. The shape maps agree.

Positions. The target $C_1 \triangleleft C_2$ over $(s_1, \lambda p_1 \cdot g(p_1, \bullet))$ has positions $\{(p_1, p_2) : p_2 : P_2(g(p_1, \bullet))\}$. On the left, $(C_1 \triangleleft \lambda_{C_2})_p$ sends $(p_1, p_2) \mapsto (p_1, (\bullet, p_2))$ and then α_p sends $(p_1, (\bullet, p_2)) \mapsto ((p_1, \bullet), p_2)$; on the right, $(\rho_{C_1} \triangleleft C_2)_p$ sends $(p_1, p_2) \mapsto (\rho_p p_1, p_2) = ((p_1, \bullet), p_2)$. They agree. $\square$

Corollary 1 (Coherence). *(**Cont**, $\triangleleft, y$) is a monoidal category with the data of Propositions 1 and 2; it is non-symmetric. By Mac Lane’s coherence theorem, Theorems 1 and 2 (with naturality of α, λ, ρ , which is immediate from the explicit formulas) imply that every formal diagram of associators and unitors commutes.*

3 Part (B): the Dirichlet tensor (**Cont**, $\otimes, y$)

3.1 The operation, its associator, and strong monoidality

Definition 2 (Dirichlet tensor).

$$C_1 \otimes C_2 = (S_1 \times S_2) \triangleleft ((s_1, s_2) \mapsto P_1 s_1 \times P_2 s_2), \quad \llbracket C_1 \otimes C_2 \rrbracket X = \sum_{(s_1, s_2)} (P_1 s_1 \times P_2 s_2 \rightarrow X).$$

Proposition 3 (Associator and unitors for $\otimes$). *The associator $\alpha_{C_1, C_2, C_3}^\otimes : (C_1 \otimes C_2) \otimes C_3 \rightarrow C_1 \otimes (C_2 \otimes C_3)$ is Cartesian reassociation on shapes and on positions,*

$$\alpha_s^\otimes : ((s_1, s_2), s_3) \mapsto (s_1, (s_2, s_3)), \quad \alpha_p^\otimes : (p_1, (p_2, p_3)) \mapsto ((p_1, p_2), p_3),$$

and the unitors are the unique maps $S_1 \times 1 \cong S_1$, $1 \times S_1 \cong S_1$ on shapes (with $P \times 1 \cong P$ on positions). The pentagon and triangle for $\otimes$ are exactly the pentagon and triangle for the Cartesian product $\times$ of sets, applied independently to the shape sets and the position sets; the braiding $(s_1, s_2) \mapsto (s_2, s_1)$ makes $\otimes$ symmetric.

Proof. $(C_1 \otimes C_2) \otimes C_3$ has shapes $(S_1 \times S_2) \times S_3$ and positions $(P_1 s_1 \times P_2 s_2) \times P_3 s_3$; $C_1 \otimes (C_2 \otimes C_3)$ has shapes $S_1 \times (S_2 \times S_3)$ and positions $P_1 s_1 \times (P_2 s_2 \times P_3 s_3)$. The reassociation isomorphisms are the components of $\alpha^\otimes$. A four-fold composite has shapes $\prod_i S_i$ and positions $\prod_i P_i s_i$ under any bracketing, and each leg of the pentagon is the corresponding product reassociation; since $(\mathbf{Set}, \times, 1)$ is a (symmetric) monoidal category, its pentagon and triangle hold, hence so do those for $\otimes$ at both the shape and position levels. Symmetry is the swap, whose hexagon is the swap-hexagon in $\mathbf{Set}$. $\square$

Theorem 3 ($\llbracket - \rrbracket$ is strong (indeed strict) monoidal for $\otimes$). *[REFUTED 2026-07-14 — see Erratum. The “indeed strict” half is false; $\llbracket - \rrbracket$ is strong monoidal for $\otimes$. The definition of $\otimes_{\text{Dir}}$ below transports the tensor along $\llbracket - \rrbracket$ and is therefore circular; the intrinsic definition is Day convolution of $(\mathbf{Set}, \times, 1)$ (Niu–Spivak arXiv:2312.00990, Prop. 3.79) and the comparison is the co-Yoneda isomorphism out of a coend.] Let $\otimes_{\text{Dir}}$ be the Dirichlet convolution on the image $\mathbf{Poly} \subseteq [\mathbf{Set}, \mathbf{Set}]$ of $\llbracket - \rrbracket$, defined on polynomial functors in their native presentation by*

$$\left(\sum_s y^{P_s} \right) \otimes_{\text{Dir}} \left(\sum_t y^{Q_t} \right) = \sum_{(s,t)} y^{P_s \times Q_t}.$$

Then the comparison $\varphi_{C_1, C_2} : \llbracket C_1 \otimes C_2 \rrbracket \rightarrow \llbracket C_1 \rrbracket \otimes_{\text{Dir}} \llbracket C_2 \rrbracket$ is the identity natural transformation, and $\llbracket y \rrbracket = \text{Id}$ is the strict unit. Consequently $\llbracket - \rrbracket : (\mathbf{Cont}, \otimes, y) \rightarrow (\mathbf{Poly}, \otimes_{\text{Dir}}, \text{Id})$ is a strict monoidal functor; in particular it is strong monoidal, and the associativity and unit coherence squares for φ commute.

Proof. [REFUTED 2026-07-14: this proof is circular — it derives “ φ is the identity” from a definition of $\otimes_{\text{Dir}}$ that was itself read off the container presentation. See Erratum. Text retained as the record of the error.] With $\llbracket C_1 \rrbracket = \sum_{s_1} y^{P_1 s_1}$ and $\llbracket C_2 \rrbracket = \sum_{s_2} y^{P_2 s_2}$, the definition of $\otimes_{\text{Dir}}$ gives $\llbracket C_1 \rrbracket \otimes_{\text{Dir}} \llbracket C_2 \rrbracket = \sum_{(s_1, s_2)} y^{P_1 s_1 \times P_2 s_2}$, whose value at X is $\sum_{(s_1, s_2)} (P_1 s_1 \times P_2 s_2 \rightarrow X)$. By Definition 2 this is *equal*, not merely isomorphic, to $\llbracket C_1 \otimes C_2 \rrbracket X$, naturally in X ; thus $\varphi_{C_1, C_2} = \text{id}$. On representables this is the familiar $y^A \otimes_{\text{Dir}} y^B = y^{A \times B}$ ($S_1 = S_2 = 1$, $P_1 = A$, $P_2 = B$). The unit object $y = 1 \triangleleft 1$ has $\llbracket y \rrbracket = \text{Id}$, the strict unit of $\otimes_{\text{Dir}}$. With φ the identity, the associativity coherence square reads $\llbracket \alpha^\otimes \rrbracket = a^{\text{Dir}}$, where a^{Dir} is the associator of $\otimes_{\text{Dir}}$; both are Cartesian reassociation of the position exponents $\prod_i P_i s_i$ (Proposition 3), so they coincide, and the unit triangles collapse since all three of φ , $\llbracket \lambda^\otimes \rrbracket$, $\llbracket \rho^\otimes \rrbracket$ are identities/Cartesian unit isomorphisms matching $\otimes_{\text{Dir}}$ ’s. $\square$

Remark 2 (Why $\otimes$ is the subtle one). [REFUTED 2026-07-14 — see Erratum. Pointwise-ness is not decided by “is it a Day convolution?”: $\times$ on $\mathbf{Cont}$ is the Day convolution of $(\mathbf{Set}, +, 0)$ and is pointwise. What decides it is whether the corepresentable splits the domain tensor ($y^{A+B} \cong y^A \times y^B$ does; $y^{A \times B}$ does not). Witness: $y^3 \otimes y^3 = y^9 \neq y^6$. The final sentence’s claim that “ φ turns out to be the identity” is also false.] For the

other three structures the target operation on $[\mathbf{Set}, \mathbf{Set}]$ is *intrinsic* to the underlying functors and pointwise or compositional: $\llbracket C_1 + C_2 \rrbracket = \llbracket C_1 \rrbracket + \llbracket C_2 \rrbracket$ (pointwise coproduct), $\llbracket C_1 \times C_2 \rrbracket = \llbracket C_1 \rrbracket \times \llbracket C_2 \rrbracket$ (pointwise product), and $\llbracket C_1 \triangleleft C_2 \rrbracket = \llbracket C_1 \rrbracket \circ \llbracket C_2 \rrbracket$ (composition). The Dirichlet convolution is *not* a pointwise or compositional operation on bare functors: $\llbracket C_1 \rrbracket \otimes_{\text{Dir}} \llbracket C_2 \rrbracket$ cannot be computed from the values $\llbracket C_1 \rrbracket X, \llbracket C_2 \rrbracket X$ alone — it reads off the polynomial *presentation* (the position exponents). This is the real sense in which $\otimes$ is subtle. That φ turns out to be the identity is precisely the statement that the container datum (S, P) is the polynomial presentation, so no comparison data is lost; the subtlety lives in the target, not in φ .

4 Computational verification

A finite-container engine (`verify_coherence.py`) implements Definition 1, Definition 2, the explicit $\alpha, \lambda, \rho, \alpha^\otimes$ and the whiskerings, and tests the pentagon and triangle as *exact* equalities of the shape/position-bijection data (μ_s, μ_p) .

- 120 randomised four-tuples (C_1, C_2, C_3, C_4) (1–2 shapes, 0–2 positions each): the sequential pentagon (Thm. 1) and triangle (Thm. 2) hold in every case, as do the Dirichlet pentagon and triangle (Prop. 3) and the strict equality $\llbracket C_1 \otimes C_2 \rrbracket = \llbracket C_1 \rrbracket \otimes_{\text{Dir}} \llbracket C_2 \rrbracket$ (Thm. 3). **[The last item is REFUTED 2026-07-14 — see Erratum. The computation tested the circular definition of $\otimes_{\text{Dir}}$, so it could only ever return “equal”; it is evidence of nothing.]**
- A deterministic branching stress case (C_1 : one shape, two positions; C_2 : two shapes, 1/2 positions; C_3 : one shape, two positions; C_4 : two shapes, 0/1 positions; four-fold composite of 400 shapes) confirms both pentagon legs agree and that the associator is genuinely non-trivial — e.g. its position component sends $(x, (u, m)) \mapsto ((x, u), m)$.
- α is confirmed to be a well-formed isomorphism: α_s a shape bijection and each α_p a position bijection.

5 Provenance and downstream

Cited in substance: Spivak, *Polynomial Functors* (2021) (substitution product, Dirichlet tensor); Niu–Spivak, arXiv:2312.00990 (Dirichlet/closed structure); full faithfulness of $\llbracket - \rrbracket$: Abbott–Altenkirch–Ghani.

MacBeth (original): the explicit container-level associator/unitors with the currying + Σ -reassociation decomposition; the flattening proof of the pentagon and the flat-tuple argument for positions; the explicit triangle; the identification of φ as the identity and the presentation-dependence diagnosis of Remark 2; the finite-container verification. **[CORRECTED 2026-07-14: the last two of these are struck. “ φ is the identity” is refuted (see Erratum), and the presentation-dependence diagnosis is both wrong and not original — Dirichlet = Day convolution is Niu–Spivak arXiv:2312.00990, Prop. 3.79, with essentially the same proof. What remains original is Part (A): the container-level associator/unitors, the flattening pentagon proof, the flat-tuple position argument, and the explicit triangle.]**

Downstream: backs Propositions 1–2 (sequential) and the Dirichlet propositions of papers/category-of-containers (PR #16) with self-contained proofs; Theorem 1 (the explicit α with transport-free components) is the raw material for the Lean pentagon/triangle target on `Composition.lean`.